

ELEC6203-Microsensor Technologies

Coursework

Mohammad Muzammil Imam (mmi2n25@soton.ac.uk)

Abstract-Present here are the results of the ELEC6203 lab sessions on the interfacing of the sensor circuits. An accelerometer sensor has been designed based on the 10g characteristics of the measurement specialities 3028 model using a resistive load circuit with 4 active elements (Wheatstone Full Bridge circuit). It has been checked that this sensor works for different values of voltage and frequency of the VCR and that it provides a sensitivity of 4.5 mV/g. It gives this desired result at $R=16\Omega$ and $\Delta R=0.072\Omega$. A differential and an instrumentation amplifier have been designed and connected to a 100Ω resistive load to amplify the sensitivity by 100 ± 5 mV/g. It has also been confirmed that the sensor performance deteriorates when the input impedance of the differential amplifier becomes equal to the output impedance of the sensor (maximum power transfer theorem). In this condition, the required output voltage decreases by half. Since the system should only pass frequencies in the range of 20-170 Hz, a band-pass filter has been designed which has a voltage roll-off of around -40dB/decade. All these are connected together, which gives an overall current draw of 135.46 μ A, which is quite good for an accelerometer sensor system. After that, the values of the connected resistors and capacitors of the BPF are increased such that they reduce the current draw. Nonetheless, the system is not able to keep the required sensitivity as the required output voltage gets reduced. All the components of the circuits are the standard values present in the Multisim directory, which may slightly differ from the theoretically calculated values.

Index terms- Band-pass filter; Differential amplifier; Instrumentation amplifier; Piezoresistive circuit, Wheatstone Full Bridge

1. What are the advantages and disadvantages of this type of accelerometer? What type(s) of interface circuit would be suitable for this device? What are the important parameters to consider?

Types of Accelerometers:

- **Piezoresistive accelerometers:** These types of accelerometer sensors use strain gauges (piezoresistors), which are often arranged in a Wheatstone bridge configuration, and are mounted on a cantilever beam. Extrinsic factors such as acceleration make the cantilever move and produce strain in the resistors that change their electrical resistance, thereby generating a proportional voltage change. These are DC-responsive, so they can measure constant acceleration like gravity as well as dynamic acceleration [1] [2].
- **Piezoelectric accelerometers:** These sensors are based on the piezoelectric effect, where acceleration on a seismic mass exerts a force on a sensing crystal, which can be made of materials like quartz or PZT. This stress then generates an electrical charge or voltage proportional to the acceleration. Typically, these are AC-responsive and do not measure constant acceleration, making them suitable for vibration and shock measurements.
- **Capacitive accelerometers:** These are generally manufactured using MEMS technology. Their seismic mass is suspended between two electrodes, forming a differential capacitor. When acceleration is applied, it causes the mass to move, changing the gap between the two electrodes and thus, proportional to the deflection, the differential capacitance changes. Typically, they are DC-responsive and are widely used in consumer electronics due to their small size and low power consumption.

Category: According to the references and the provided datasheet, the 3028 accelerometer by Measurement Specialities falls in the category of piezoresistive accelerometers.

Advantages of piezoresistive accelerometers:

- **Measures static acceleration:** These devices are DC responsive since they are capable of measuring acceleration down to 0 Hz, including static forces like gravity and constant motion.
- **Excellent for high amplitude shock testing:** A very wide bandwidth and high dynamic range make this sensor ideal for high frequency, short duration events like crash and weapons testing.
- **Robust:** These types of devices can be gas or fluid damped, which prevents them from reaching their internal resonant frequency and provides greater resistance to high gravity shock.

Disadvantages of piezoresistive accelerometers:

- **Temperature sensitivity:** These are very sensitive to temperature variation, leading to thermal zero shift and thermal sensitivity shift, which often requires internal or external circuitry for temperature compensation.
- **Lower sensitivity:** They generally have lower sensitivity compared to other types of accelerometers, resulting in a lower output signal level, which makes it necessary to use a high-gain interface circuit.
- **High cost:** It is also more expensive compared to other types of accelerometers, making it less common in mass-market consumer applications [3].

Interface circuit

The 3028 model is based on the Wheatstone bridge circuit, which produces a low-level differential mV/g output. I have used a Resistive load circuit with four active elements, arranged in a full-bridge Wheatstone circuit, so that when an external force, i.e. voltage, is applied, the resistance of the two diagonal arms decreases while the other two arms increase with respect to the base connected resistance. After that, I have calculated the ΔR so that it matches the required sensitivity value [4][5].

This interface circuit includes [6]:

- **Excitation voltage source:** Stable, Low-noise 5 V DC excitation to power the sensor.
 - **Differential or Instrumentation Amplifier:** It amplifies the output voltage from the sensor for further filtering purposes and to reject the common-mode noise.
 - **Band-pass filtering:** It removes the unwanted ripples from the required output response in a frequency range.
2. Find the (simplified, ideal) equivalent circuit of the accelerometer, and simulate it at a reasonable range of frequencies and amplitudes to verify the behaviour of its differential outputs. You are encouraged to do this in NI Multisim. You should set up the simulation so that the input vibration amplitudes and frequencies can be easily varied. You should assume that the resistances are precisely matched, and the sensitivity is in the centre of the range specified in the datasheet.

It is given that the sensitivity is 4.5 mV/g for an applied DC voltage of 5V.

$$S = 4.5 \frac{mV}{g} \text{ (Centre of the range specified in the datasheet)} \quad V_e = 5 V$$

$$S = \frac{\Delta R}{R} = \frac{V_o}{V_e} \quad (i) \quad V_o = V_e * S \quad (ii)$$

$$V_o = 5 * 4.5 * 10^{-3} V = 22.5 mV$$

Now that equations (i) and (ii) give us the relation between the output voltage and the required sensitivity, we have to select a base resistor value and scale it so that we get the required output voltage and calculate ΔR from that [7][6].

Values of VCR:

$$V_{RMS} = 0.7071 * 5$$

$$Frequency = 100\text{ Hz}$$

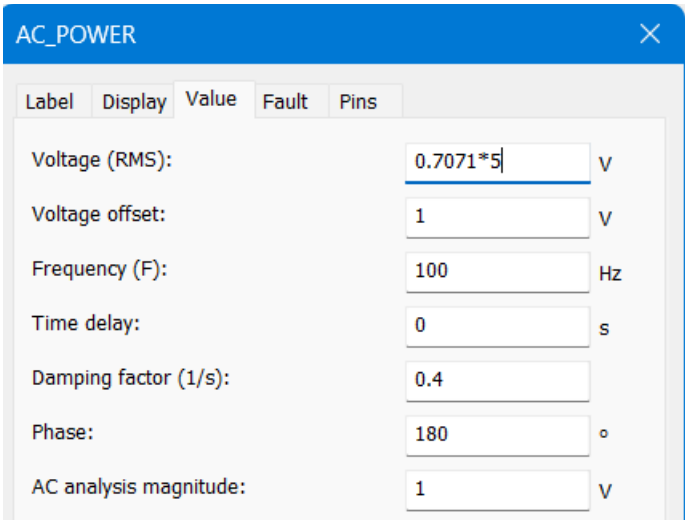


Figure 1 Input parameters of the VCR

Input impedance of the sensor = 3.5kΩ

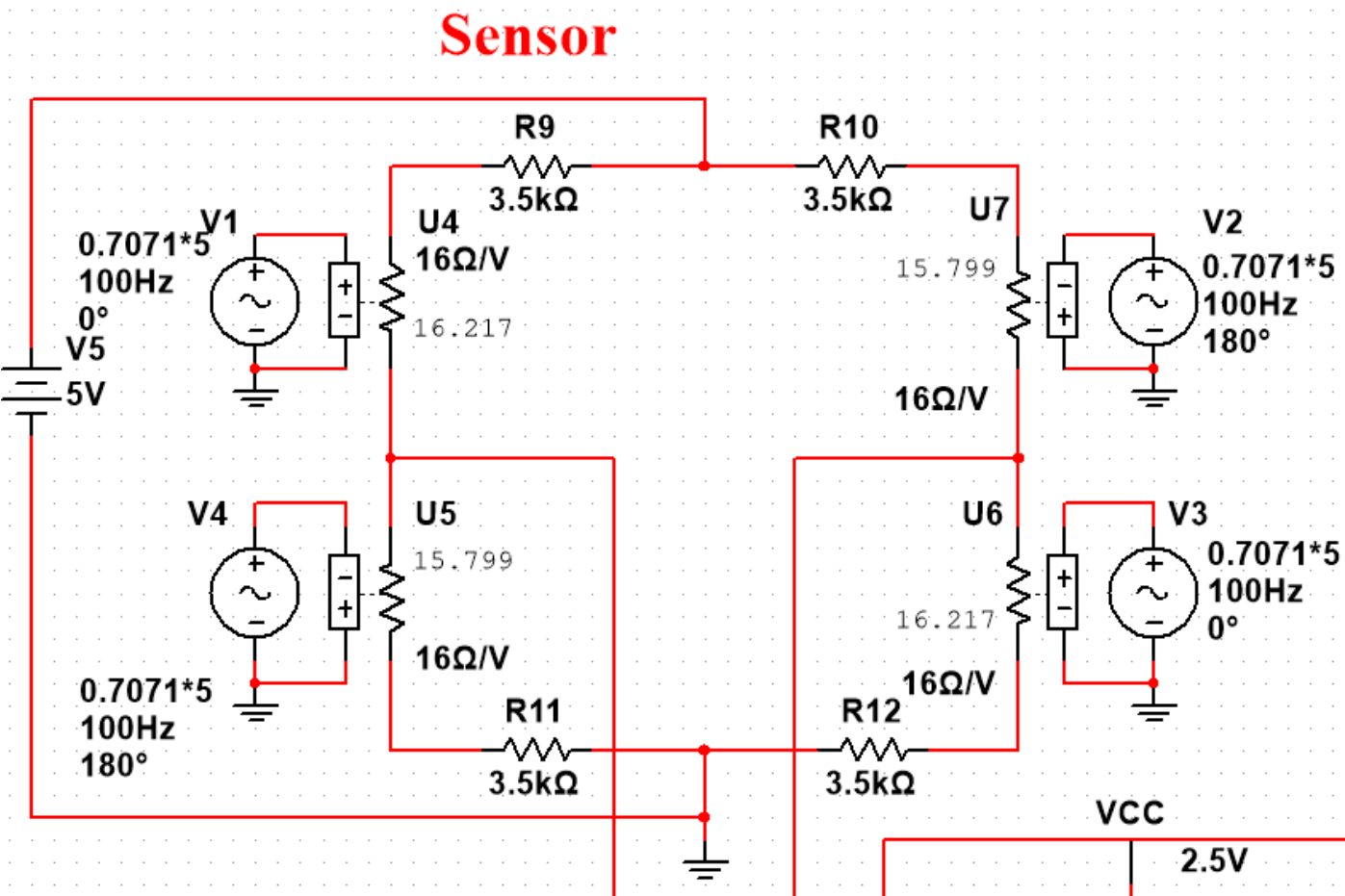


Figure 2 Equivalent sensor circuit

This is a Full Bridge Wheatstone circuit in which the values of two of the diagonal VCRs increase while the other two decrease when an external voltage is applied to it, which is the typical working principle of the piezoresistive sensor.

Table 1 Scaling resistance values

Resistance (Ω)	V_o (mV/g)
16	22.5
50	71.429
100	141.679
160	225.835
500	714.286
1k	1.429

Values extracted from the circuit:

$R + \Delta R = 16.072 \, \Omega$ $R - \Delta R = 15.028 \, \Omega$

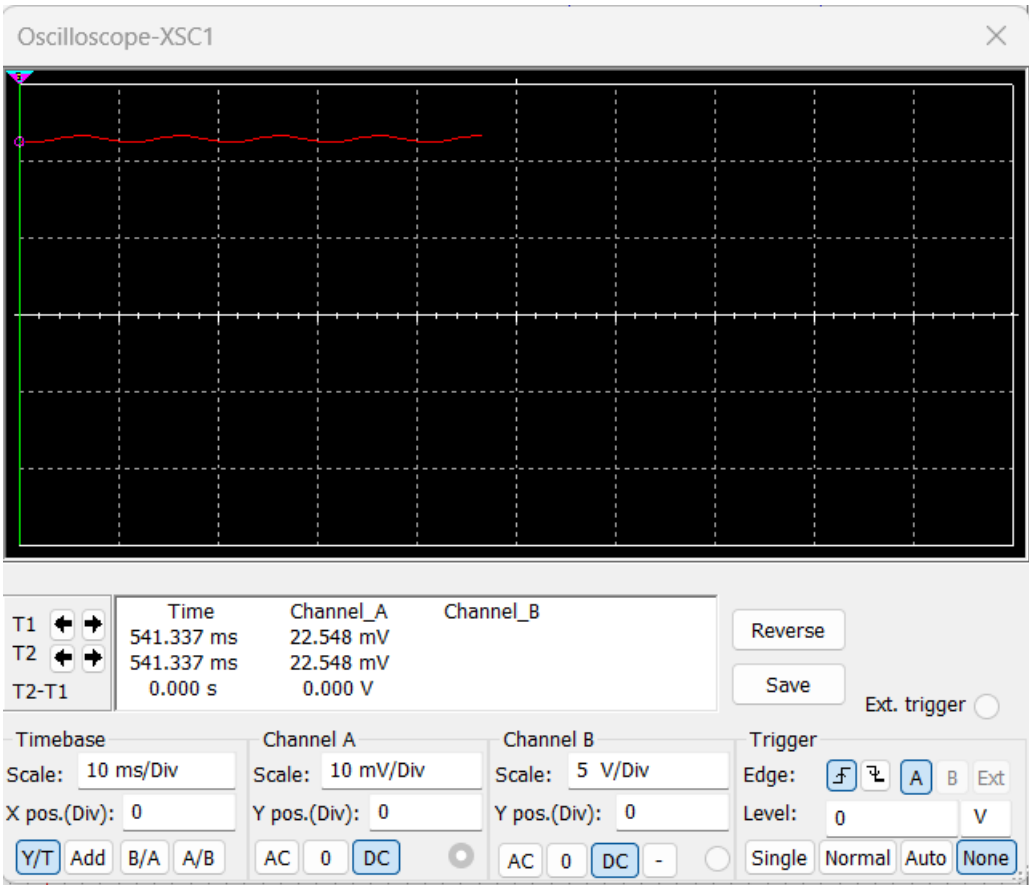


Figure 3 Oscilloscope output of the sensor

$V_o = 22.548 \frac{mV}{g}$

Therefore, $\Delta R = 0.072 \, \Omega$. At these R and ΔR values, we achieve the approximate output voltage value of 22.548 mV/g.

Table 2 Values of output voltage at different VCR parameters

V_{RMS} (V)	Frequency (Hz)	V_o (mV/g)
0.7071*5	100	22.548
0.7071*5.5	200	22.734
0.7071*4.5	300	22.682
0.7071*4	300	22.670
0.7071*2.5	400	22.756

Conclusion:

It can now be concluded that the simulated circuit works for the differing parameter values of the VCR.

The frequency of the VCR doesn't have much effect on the output voltage. The input voltage of the VCR is mostly responsible for the variation in the output voltage, and even for that, it can be said that the variation occurs only in a very small range.

3. Connect the outputs from the accelerometer to a differential amplifier, and the amplifier output to the 100Ω resistive load. You should choose appropriate components to increase the sensitivity to $100 \pm 5 \text{ mV/g}$, and explain why you chose this op-amp and the other components. Analyse and document the circuit's performance.

Given, $S = 100 \pm 5 \frac{\text{mV}}{\text{g}}$ and in this case $V_e = 22.548 \text{ mV}$ as it is the output of the accelerometer.

From equations (i) and (ii),

Now, V_o of the amplifier with a 100Ω resistive load should be in the range of

$$V_o = 105 * 22.548 = 2.367 \text{ V}$$

$$V_o = 95 * 22.548 = 2.142 \text{ V}$$

We now have to select the R_1 and R_2 values of the Differential amplifier so that it takes the input voltage from the sensor and amplifies it to the required output voltage [6].

We know that the differential gain of the amplifier is given by,

$$m = A_{diff} = \frac{R_2}{R_1}$$

To get the required output voltage in the range of 2.142 V to 2.367 V, let's assume $R_2 = 110\text{k}$ and $R_1 = 1\text{k}$. Therefore, it gives us a gain of 110.

$$m = A_{diff} = 110$$

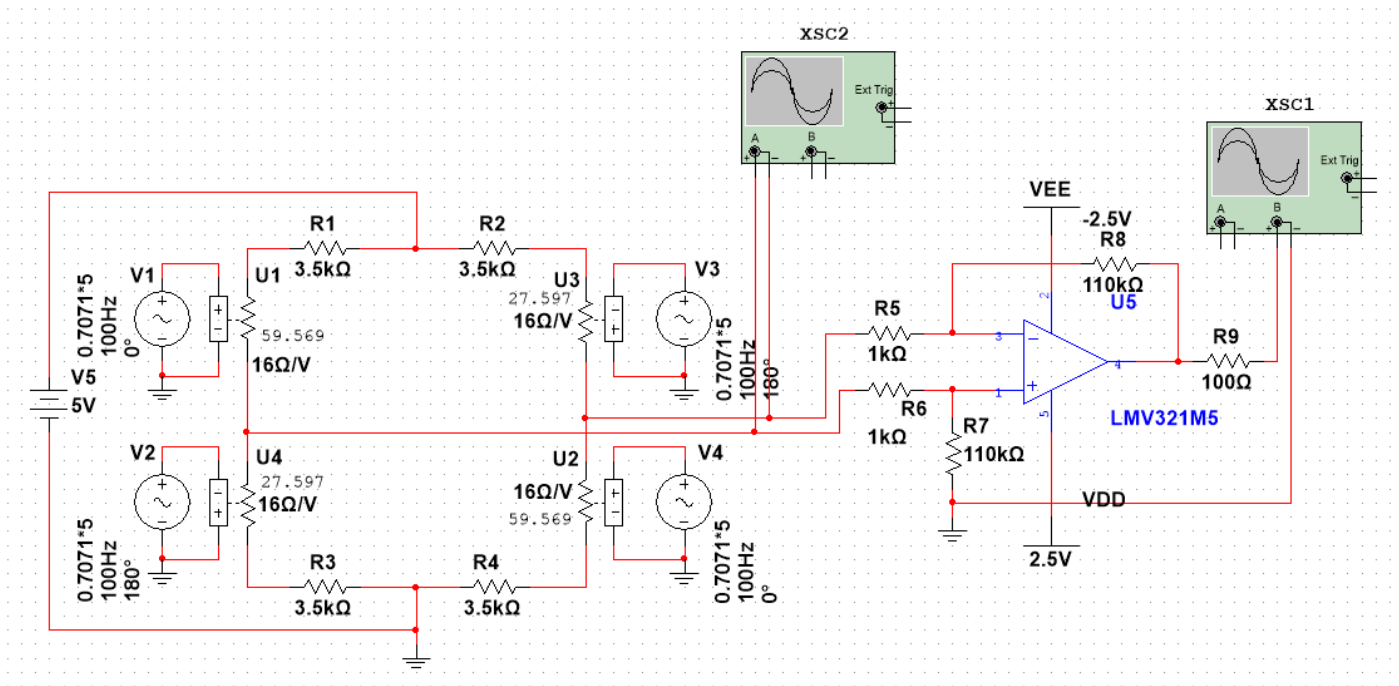


Figure 4 Sensor with a differential amplifier

Output of the differential amplifier is:

$$V_o = m(E_1 - E_2)$$

where $E_1 = 2.51\text{V}$ and $E_2 = 2.49\text{V}$ and, since $R_5=R_6=1\text{k}\Omega$ and $R_7=R_8=110\text{k}\Omega$, $m = \frac{R_7}{R_6} = 110$ therefore,

$$V_o = 2.095\text{V}$$

which is quite close to the required value of

$$V_o = 2.142\text{V}$$

Reasons for choosing LMV321M5 OPAMP:

- It is a general-purpose OPAMP which has a single, low voltage rail to rail output and is suited to work on a 5V power supply [8].
- The rail-to-rail (RRO) output stage means that its output voltage can swing very close to both the positive supply (V_{CC}) and the negative supply (V_{EE}), which ensures that the maximum required signal is properly preserved.
- The sensor circuit is designed to run from a single 5V source, and LMV321M5 is specifically designed to run from a low supply voltage range of 2.7V to 5.5V, making it compatible with the required circuit.
- “M5” designator refers to the SOT-23-5 package, which is a very small, surface-mount package, making the circuit space-efficient and suitable for portable and compact designs.

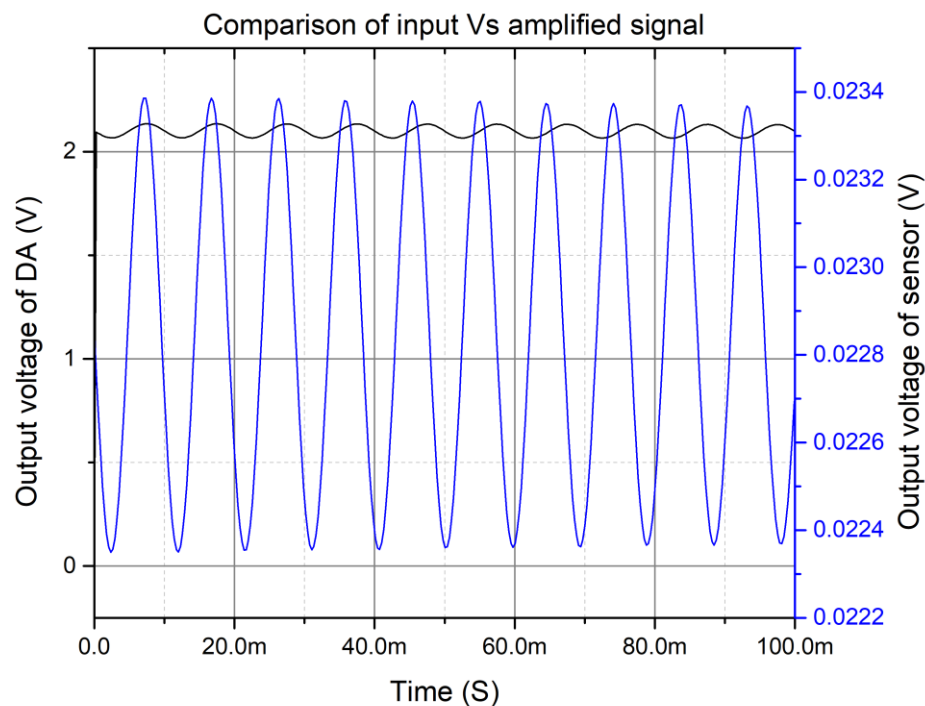


Figure 5 Oscilloscope output of the differential amplifier

Discussion of how the choice of these components affected the results:

Choice of OPAMP:

To design the differential amplifier with these specifications, I had to choose an OPAMP which supports:

- Low power signal output
- Has a single rail-to-rail output circuit
- A power supply of 5V
- It must not be a specialised OPAMP
- And it must not be an already made amplifier circuit module.

These conditions led me to use the LMV321M5 general purpose OPAMP, which provided accurate but not so precise results [8].

Choice of the resistor value:

The resistances of the DA are of 110k Ω and 1k Ω , which are not the precise calculation for the required gain. I have used these resistances because they are generally available in the market. Using these minutely changed the desired gain and the output voltage of the amplifier from 2.142 V to 2.095 V.

4. Evaluate the effect of matching the input impedance of the differential amplifier with the output impedance of the accelerometer on system performance. You should present this both theoretically (for an ideal circuit) and with simulation.

The most affected parameter in a sensor to amplifier system is the actual voltage gain measured by the amplifier stage, which is determined by the voltage divider effect at the amplifier's input [2].

Assuming,

V_{CC} be the ideal output voltage of the accelerometer.

R_{OUT} be the output impedance of the accelerometer.

R_{IN} be the input impedance of the differential amplifier circuit.

V_{IN} be the actual voltage signal delivered to the amplifier input pins.

$$V_{IN} = V_{CC} \left(\frac{R_{IN}}{R_{OUT} + R_{IN}} \right) \quad (\text{iii})$$

Case 1: Unmatched impedance

For voltage sensing applications, the goal is to transfer the maximum voltage signal from the sensor to the amplifier, which requires impedance bridging.

$$R_{IN} \gg 10 \cdot R_{OUT}$$
$$\frac{100R_{OUT}}{R_{OUT} + 100R_{OUT}} = 0.99$$

The input impedance should be much larger than the output impedance, in the general case, more than 100 times so as to reduce the common mode gain and to reduce the attenuation. Because of this, $V_{IN} = V_{CC}$, the amplifier sees almost the full sensor voltage, which is ideal for voltage amplification, as the signal is preserved.

Case 2: Matched impedance

$$R_{OUT} = R_{IN}$$

Impedance matching is required when the goal is Maximum Power Transfer, which is usually used for high frequency RF circuits and transmission lines. This is not the goal in low frequency sensor signal conditioning.

$$\frac{R_{OUT}}{R_{OUT} + R_{OUT}} = \frac{1}{2}$$

Because of the matched impedance, the voltage signal delivered to the amplifier is exactly halved, and the gain is reduced by 50%, which significantly degrades the sensor performance by losing half the signal strength.

Simulating a matched impedance circuit:

It is given the datasheet that the typical input impedance of the sensor circuit is 3.5 k Ω .

Initially, both the inverting and non-inverting inputs of the DA were of 1 k Ω each. To make a matched circuit, both the input impedances of the DA should be half the sensor impedance.

$$R_+ = R_- = \frac{R_{sensor}}{2}$$

Therefore,

$$R_+ = R_- = 1.7 \text{ k}\Omega$$

Then the output voltage $V_o = 975.011 \text{ mV}$, which is approximately half the voltage of the unmatched circuit, indicating a substantial amount of signal loss, which is a great disadvantage for a sensor system.

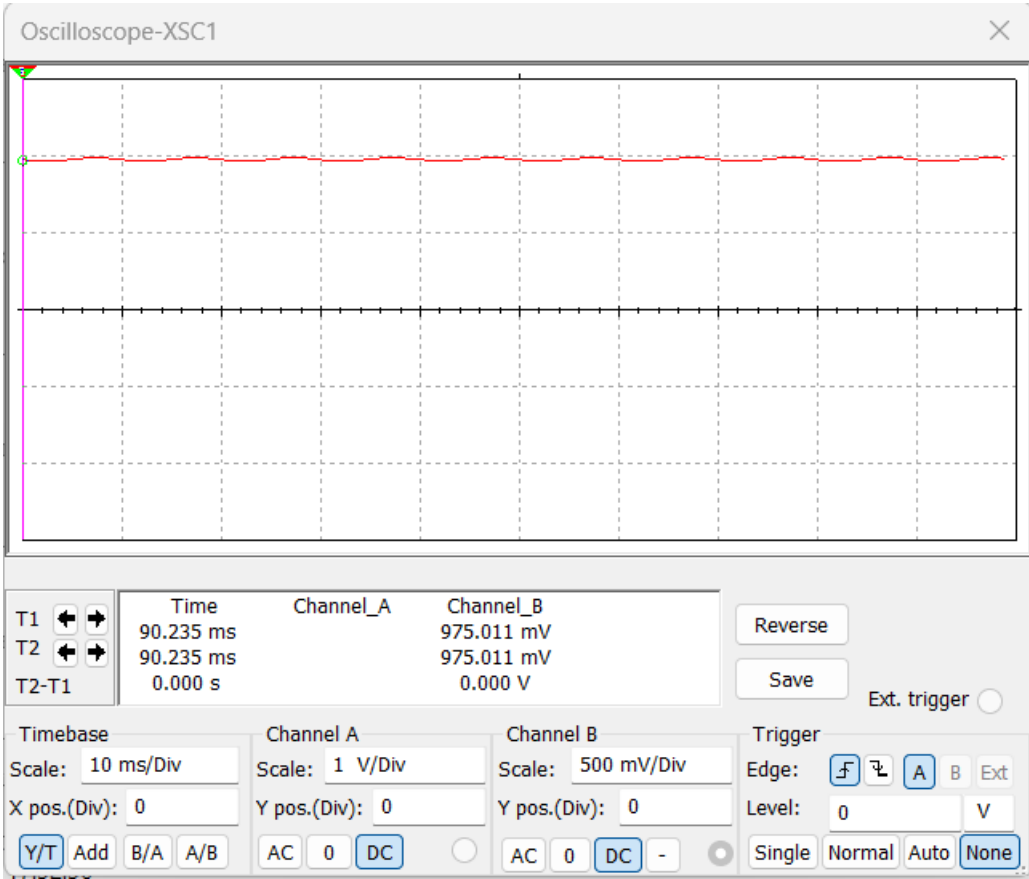


Figure 6 Matched circuit output voltage

5. Replace the differential amplifier with an instrumentation amplifier (you should construct this from individual op-amps, rather than using a dedicated instrumentation amplifier component). Again, choose appropriate components. Analyse and comment on the performance of the circuit. How does this performance differ from the differential amplifier?

Generally, an instrumentation amplifier is considered better than a differential amplifier for applications requiring high precision measurements and signal conditioning. This happens because of the following key advantages of the instrumentation amplifier [1][6]:

- **Extremely high input impedance:** A basic DA has a relatively low input impedance, and the input impedance observed at each terminal is often unequal, which can cause overloading, especially with high impedance sensors, leading to signal attenuation and measurement errors.
- **Eliminates loading effects:** The INA provides buffer amplifiers at its input stage, which present extremely high and equal input impedance at its non-inverting terminal, virtually eliminating loading effects on the signal source.
- **Easy gain adjustment:** The gain of an INA can be precisely set by adjusting the value of R_G , which simplifies design and calibration significantly.

$$A_{diff} = 1 + \frac{2R_1}{R_G}$$

- **Superior CMRR:** Common Mode Rejection Ratio is the measure of an amplifier’s ability to reject a voltage common to both input lines while amplifying the desired differential signal. The high CMRR of an INA is critical for a better SNR and a more stable output.

Parametric values of the 3 OPAMP INA:

$$R_1 = 100\Omega \quad R_2 = 10k\Omega \quad R_3 = 340k\Omega \quad R_G = 2\Omega$$

Now,

$$A_{diff} = 1 + \frac{2 * 100}{2}$$

$$A_{diff} = 101$$

$$Total\ gain = A_{diff} * \frac{R_3}{R_2}$$
$$Total\ gain = A_{diff} * \frac{340}{10} = 3434$$

$$V_o = 3434 * 22.583\ mV$$
$$V_o = 7.75\ V$$

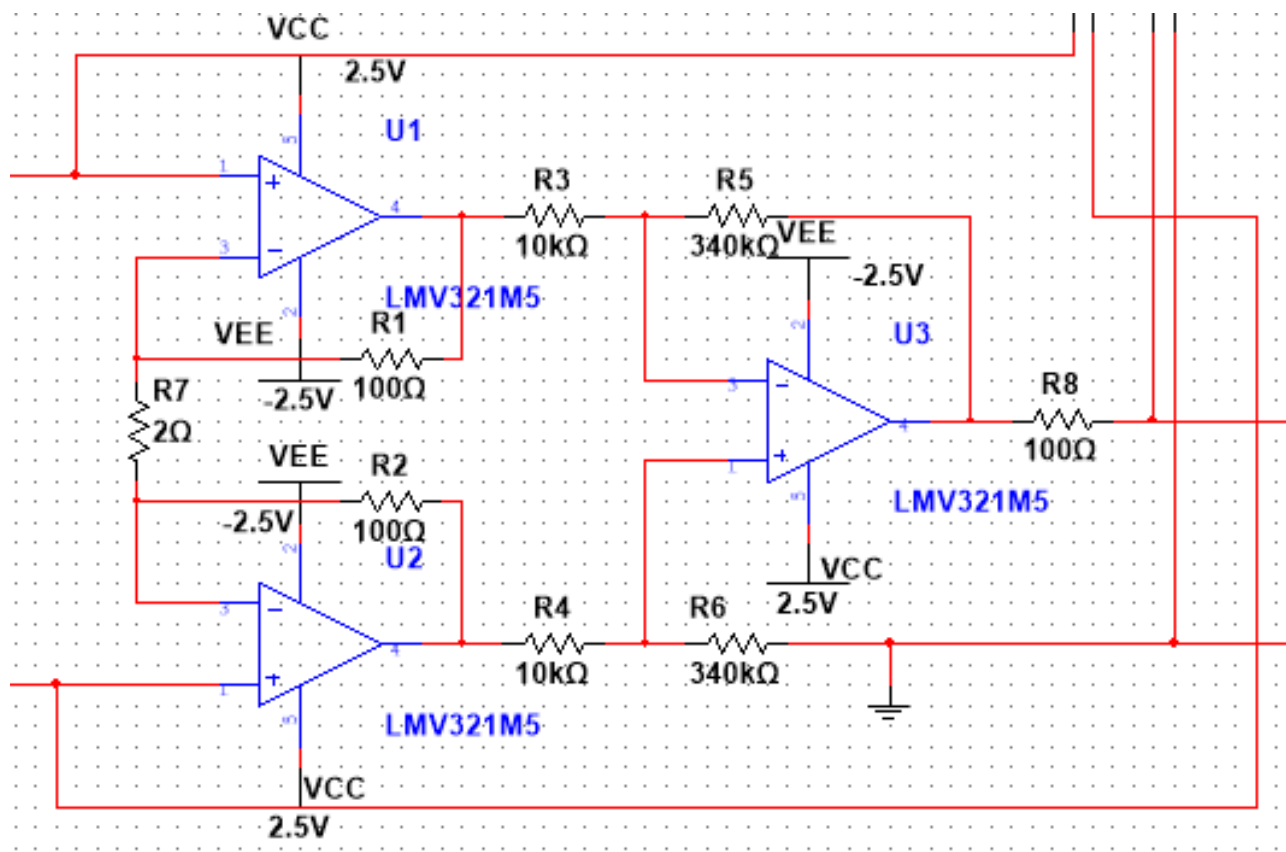


Figure 7 Sensor circuit with 3 OPAMP instrumentation amplifier

Observation:

The value of R_G is kept very small compared to R_1 so as to increase the differential gain of the amplifier.

It can also be observed that the theoretical output voltage is 7 V, which is more than the supplied voltage, and because of that, the output hits the power supply ceiling and the voltage clips at around 2.5 V. Because of this property of the incorporated opamp, we get the desired sensitivity and the output voltage.

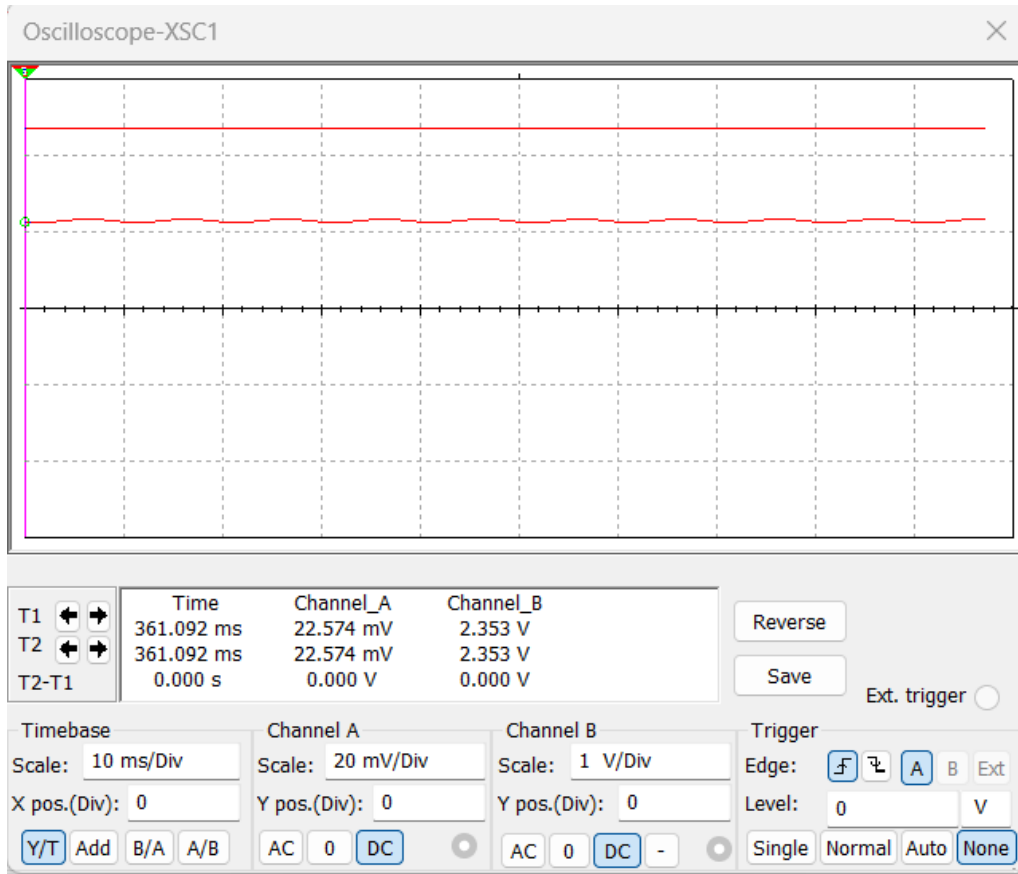


Figure 8 Oscilloscope output of the instrumentation amplifier

6. The control system is only interested in vibration in the range 20-170 Hz; you should add an active filter to your circuit so that unwanted frequencies are removed. The priority is to have as little amplitude distortion as possible in the pass-band. You should achieve a roll-off of around -40dB/decade. You are expected to explain your design, please do not use automated filter design tools for this. Present tests for the complete system, at a range of input frequencies to confirm that the system performs as designed.

According to the requirements of the circuit, a band-pass filter is needed for this operation [8].

A band pass filter incorporates a high pass filter, an amplifier, and a low pass filter connected together in such a manner that it attenuates all the unwanted frequencies and only passes the required band.

It is given that the required frequency band is 20-170 Hz, so the band-pass filter should pass frequencies higher than 20 Hz and smaller than 170 Hz [6][4].

It is also given that we need to have a gain roll-off of around -40dB/decade. For that, we need a second-order filter.

Designing an LPF and HPF:

I have chosen a Sallen-Key filter for the design, therefore,

$$f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \quad f_L = 20 \text{ Hz} \quad f_H = 170 \text{ Hz}$$

Then R comes out to be,

$$R = \frac{1}{2\pi C f_c} \quad (\text{iv})$$

Assuming the value of capacitance for HPF and LPF: $C_L = 100 \text{ nF}$ $C_H = 470 \text{ nF}$

Inputting all the values of resistances and capacitances in equation (iv), we get,

$$R_L = 9.36 \text{ k}\Omega \quad R_H = 16.93 \text{ k}\Omega$$

Using standard values as $R_L = 9.1 \text{ k}\Omega$ $R_H = 16 \text{ k}\Omega$

For the amplifier stage, the values of the resistors are taken as 40k and 10k. so when the signal at the input goes through the HPF, the unwanted frequencies get attenuated, then it is further amplified, and lastly passes through the low pass filter stage. The values of all the components are selected so as to match the practical use in everyday life.

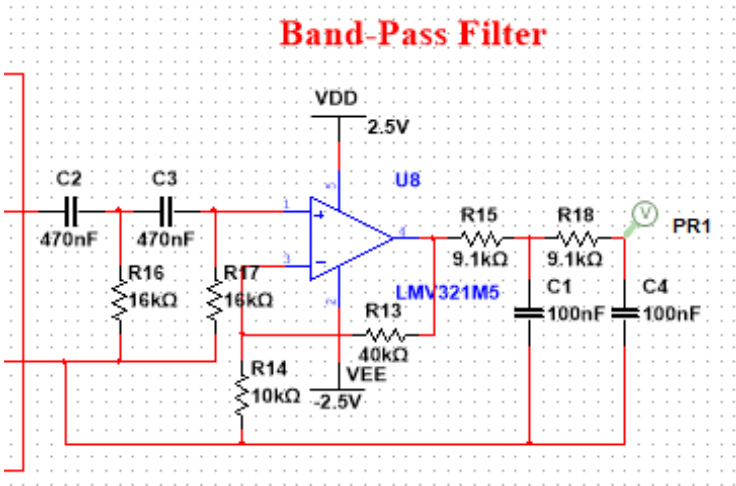


Figure 9 Band-pass filter design

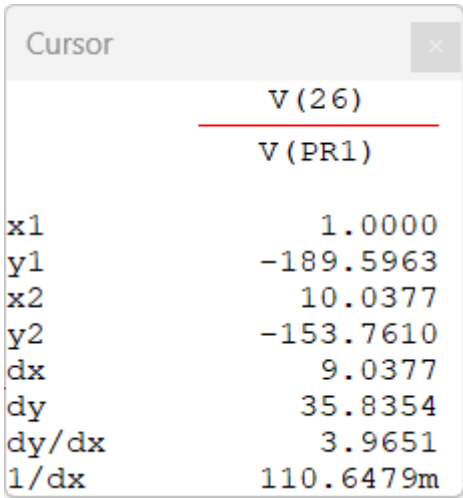


Figure 10 Cursor value of AC sweep

Observation:

It can be found from Figure 8 that when we scale the cursor from 1 Hz to 10 Hz, the change in the magnitude is around -35 dB/decade, which is quite close to the required value of -40 dB/decade. The reason for this discrepancy may be the use of standard values of the resistances and capacitances rather than the calculated theoretical values.

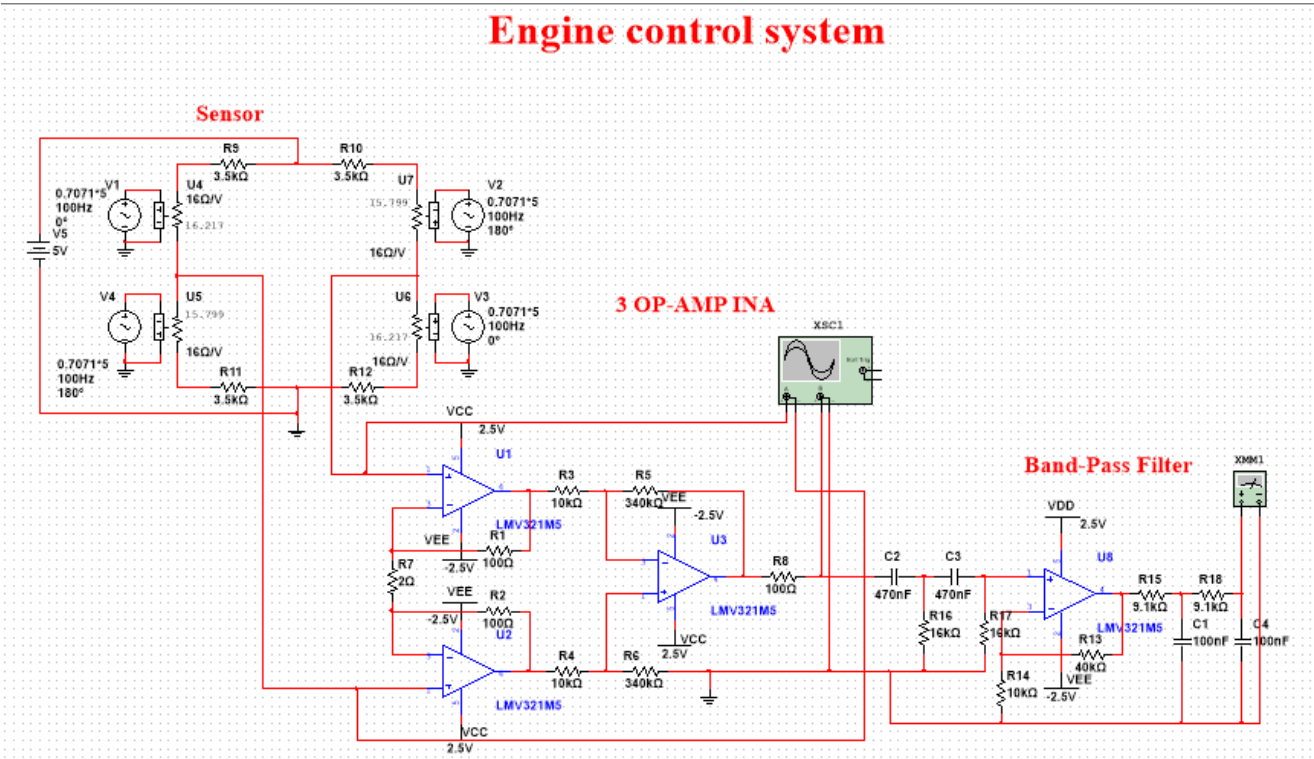


Figure 11 Engine control system

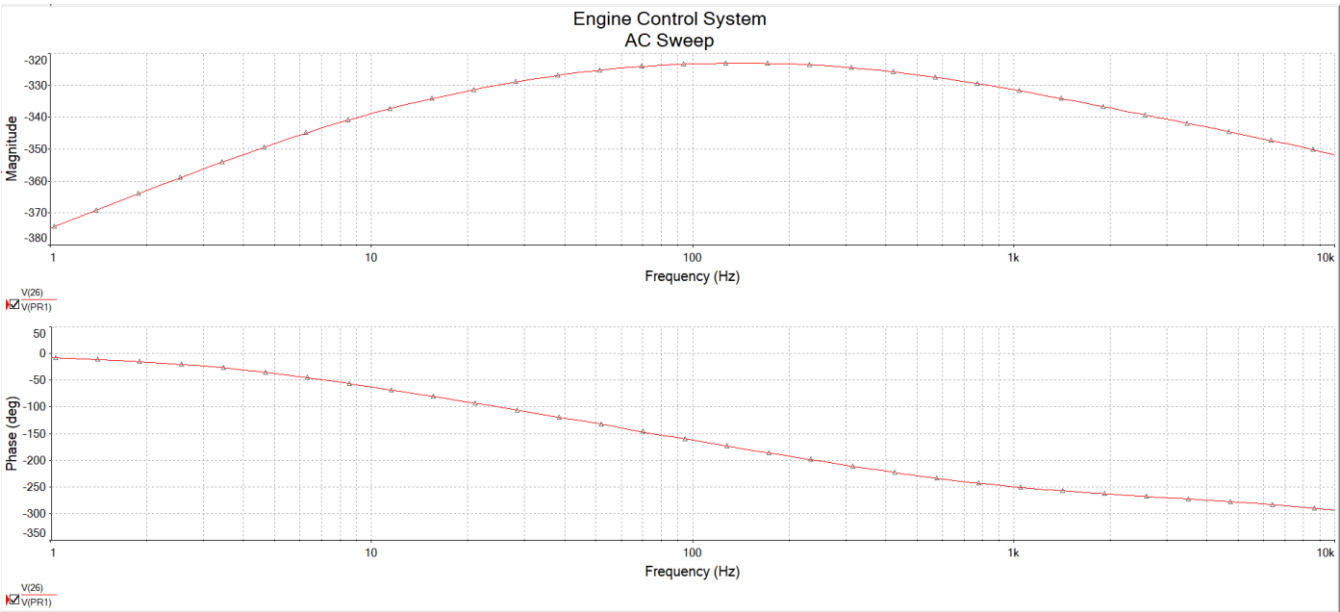


Figure 12 AC Sweep of the Engine Control System

Cursor	
V (26)	
V (PR1)	
x1	169.7158
y1	-323.1781
x2	19.9501
y2	-331.7984
dx	-149.7657
dy	-8.6203
dy/dx	57.5585m
1/dx	-6.6771m

Figure 13 Working frequency range

Observation:

AC sweep measures the phase and magnitude response of the system in logarithmic, decibel, and linear domains, which helps us calculate parameters like the gain roll-off, 3 dB frequency, etc.

It is also evident that the filter has an approximately flat output in the working frequency range.

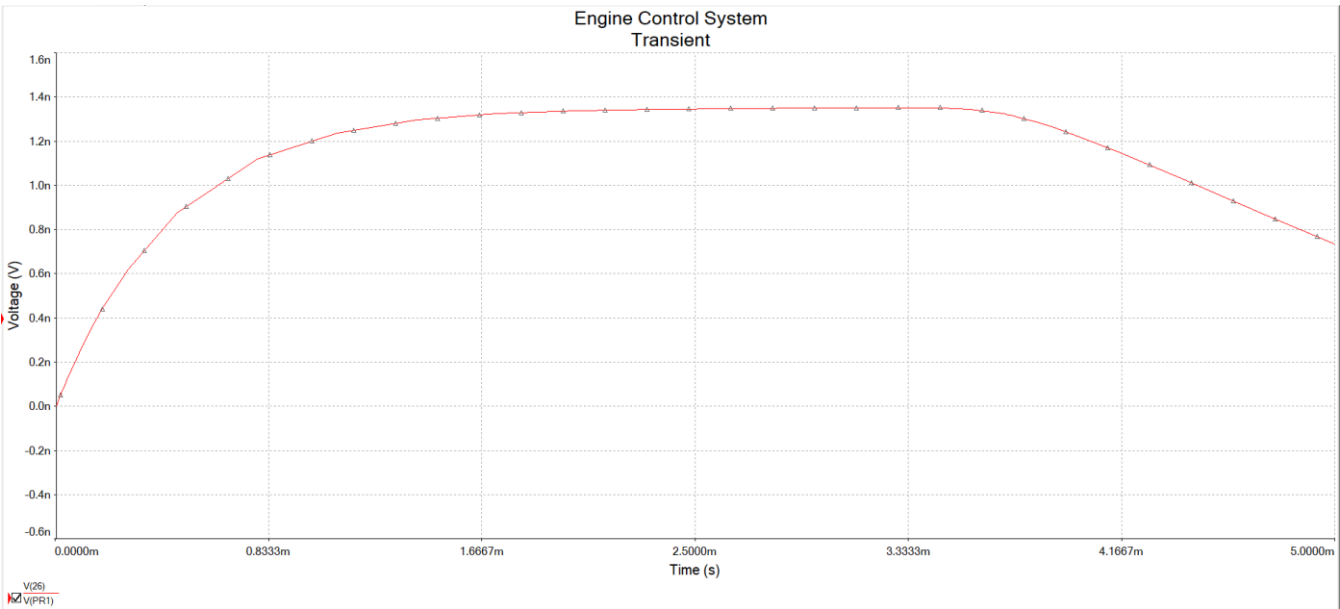


Figure 14 Transient response of the Engine Control System

Observation:

Figure 11 shows a typical band-pass filter response

7. Evaluate the overall current draw of the system. Improve this, but maintain the required sensitivity. What did you change? How has it affected performance?

The current drawn from each component is measured separately by using an ammeter.

Current drawn by:

- i. Sensor: 17.79 μA
- ii. INA with resistive load: 0.021321 μA
- iii. Band-pass filter: 117.652 μA

The overall current draw of the system is the total sum of all the output nodes of its components, and it is equal to 135.46 μA .

This value of current draw is excellent for most modern battery powered electronics, and it comes in the category of high performance low power devices, because it is low enough to run a device for months or even years on a small battery, which is beneficial for an accelerometer sensor.

How can we decrease the current draw while keeping the sensitivity same?

Supply voltage optimization: Reducing the rail voltage to the minimum operating voltage of the components.

Impedance and capacitance scaling: All the resistor values in the band pass filter have been scaled up by 10, and all the capacitor values are scaled down by 10, which maintains the centre frequency and gain because the RC time constant remains constant. This reduced the current flowing through the filter by 90%.

Trade-off: The thermal noise of the system increases and makes the circuit more sensitive to stray capacitances, because of which there is a slight decrease in the simulated output voltage.

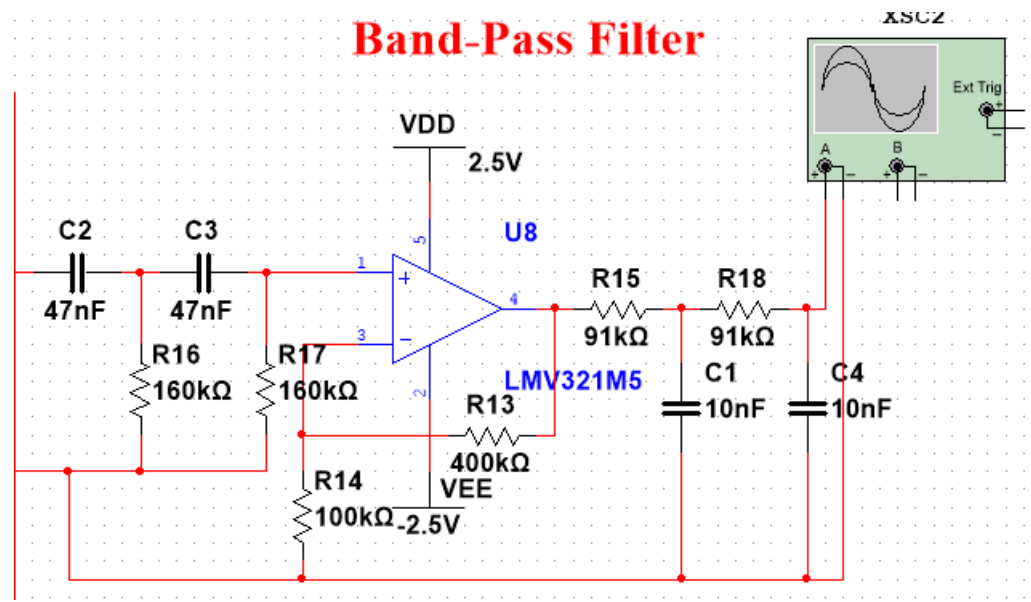


Figure 15 Optimized BPF

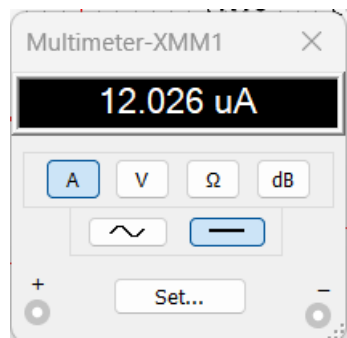


Figure 16 Current draw of BPF after optimization

Observation:

This result suggests that when the input impedance of a device is increased, it results in the reduction of the total current drawn, as there is a reduction in the amount of leakage current. This also results in the decrease in sensitivity as the required output voltage decreases.

Conclusions

In summary, an engine control system based on the Measurement Specialities 3028 accelerometer is designed on the basis of a full bridge Wheatstone circuit, which has a sensitivity of 4.5 mV/g. The desired output voltage could be measured when across a base VCR of 16 Ω and ΔR of 0.072 Ω, the output signal of which is then amplified and filtered out for accurate performance. The effect of the voltage and frequency change of the VCR on the performance of the sensor proves that it is working under varying conditions. The final results indicate that the device has an approximate voltage roll-off of around -40dB/decade, and it has a very low current draw, which is further reduced by increasing the input impedance of the system. The demerit of which, is the reduction of the required output voltage, which results in the loss of sensitivity.

References

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